SOFC stack - first data file (30 October 2025)

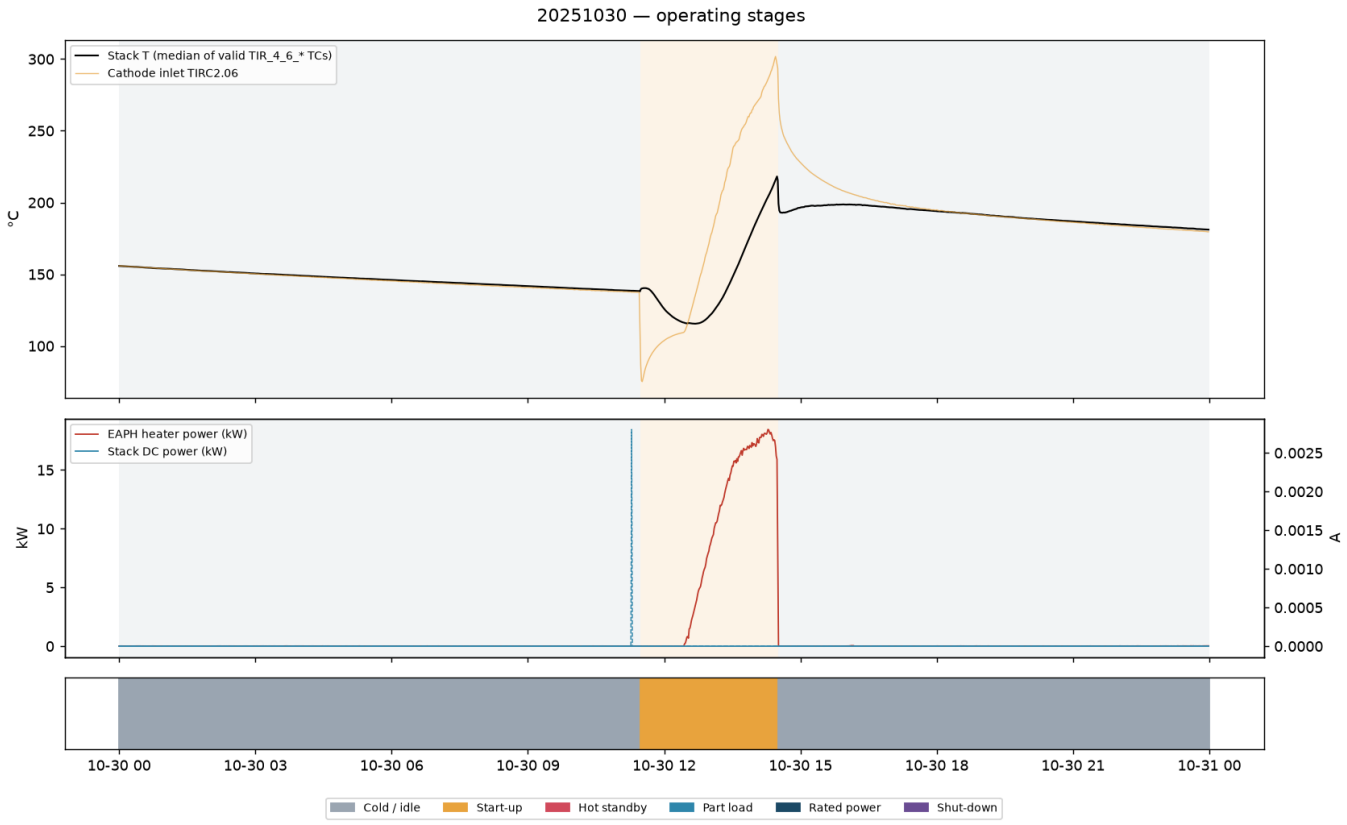
Separating the day into stages and operations

Contents: 1. The answer - 2. How each minute is labelled - 3. Which sensors - 4. Results and next steps

1. The stack made no power that day. It tried to warm up once, then stopped.

Using the rules in the operating-modes guide, the day splits into **two stages** and **four operations**. For most of the day the stack sat warm and idle. For three hours around midday it went through a start-up attempt. That attempt ended at about 216 °C, far short of the ~580 °C needed before the stack can make power.

Why it matters: this day is a test of the warm-up equipment, not a working day. Studies of power output, efficiency or wear should leave it out, or treat it as start-up data only.



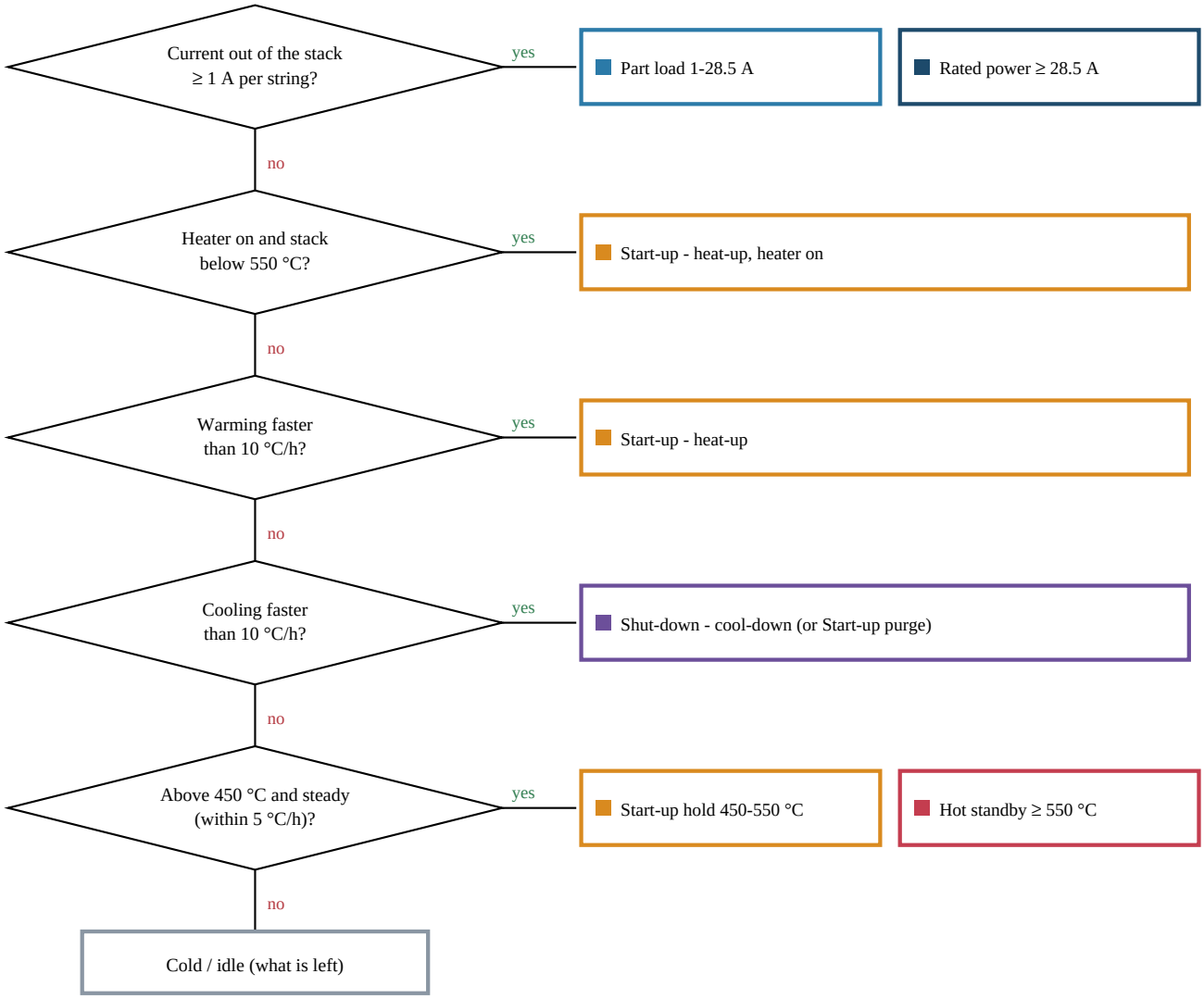
Top: stack temperature (black) and air entering the stack after the heater (orange). Middle: heater power. Bottom: stage colour band.

Stage	What was happening	From - to	Stack °C	Per hour
Cold / idle	Nothing on; slow loss of heat	00:00-11:29	156 → 138	-1.6 °C
Start-up	Air fan on, heater off: purge	11:29-12:27	139 → 116	-19 °C
Start-up	Electric air heater up to 18 kW	12:27-14:31	116 → 216	+39 °C
Cold / idle	Air and heater off together; settling	14:31-16:10	199 → 199	0 °C
Cold / idle	Slow natural cooling	16:10-24:00	198 → 181	-2.2 °C

If you remember one thing: 30 October 2025 is a start-up attempt abandoned at about 216 °C. The stack never produced power.

2. Every minute of data answers five yes/no questions

The operating-modes guide defines five stages: Start-up, Hot standby, Part load, Rated power and Shut-down. It describes each one by two things: is the stack producing power, and what is its temperature doing. Each minute goes past five gates in a fixed order and drops out at the first gate that says "yes".



Gate	Threshold	Source
Power flowing	≥ 1 A per string	Guide §1
Rated, not part load	≥ 28.5 A per string	Guide §1 (30 A rated; 95 % allows for noise)
Warming / cooling	± 10 °C per hour	Guide §1
Steady	within ± 5 °C per hour	Guide §1
Hot standby	450 °C / 550 °C	Guide §1 and §4.3 (the ~510 °C hold is Start-up)
Load ramping	± 0.05 A per minute	Guide §4.4
Heater on	> 0.2 kW	Added (off-state noise below 0.05 kW)

Three rules that look at neighbouring blocks

Cooling before a heat-up is a purge. The air fan starts before the heater, so cool air lowers the stack temperature. If a heat-up follows and the stack has not been hot since it was last cold, the block is Start-up (air purge). **Load falling to zero is Shut-down** step 1. **A heat-up ends when the heater stops**, at the

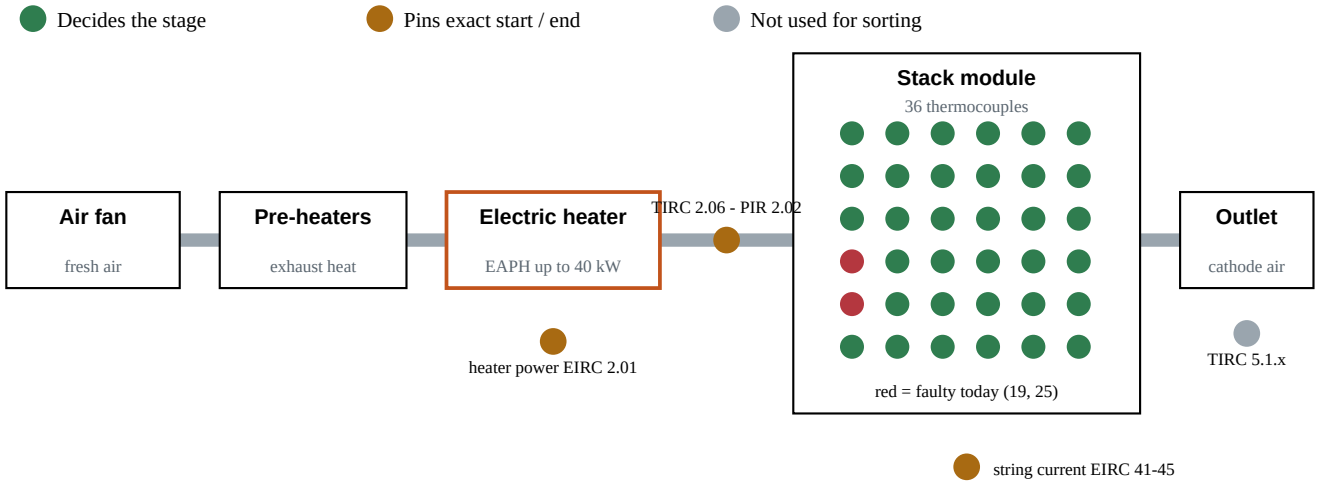
exact minute. Blocks shorter than 15 minutes merge into their neighbour.

One label the guide does not have. On this day the stack spent 21 hours sitting warm with nothing switched on. None of the five stages fits, so these hours are labelled Cold / idle.

If you remember one thing: power is checked first, then the heater, then the temperature trend. The first "yes" decides the stage.

3. Use the 36 thermocouples inside the stack. Take their middle value.

The data has 112 temperature columns. The guide defines each stage by the **stack temperature**, and only the 36 thermocouples in the stack module (TIR 4.6.1 to 4.6.36) measure it directly. The others sit on pipes, heaters or the exhaust. A thermocouple is a small wire probe that turns heat into a tiny voltage.



Why the middle value (median), not the average. Two probes were broken on this day. One was stuck at 1300 °C, a typical sign of a cut wire. The other read room temperature while the stack was at 150 °C. The median ignores them. The script also drops any probe that sits more than 75 °C from the others all day.

Probe	Lowest	Typical	Highest	Verdict
TIR 4.6.19	1300 °C	1300 °C	1300 °C	Excluded - stuck
TIR 4.6.25	21 °C	23 °C	30 °C	Excluded - reads room air
TIR 4.6.1-4.6.8	90 °C	155 °C	286 °C	Kept - run hot during heat-up
Other 26	114 °C	152 °C	227 °C	Kept

Role	Signals	Why
Decides the stage	TIR_4_6_1 … 36 (median); EIRC_41…45_CurrentMon	Stack temperature and its rate; current ≥ 1 A means power
Pins the boundaries	EIRC_2_01 heater power; TIRC_2_06 inlet air; PIR_2_02 inlet pressure	React within a minute of a switch; TIRC_2_06 is also the guide's 580 °C inlet check
Not used for sorting	TIRC_5_1/5_2 outlets; TIR_3_* fuel line; TIR_7_* exhaust; TIRZ_8_* safety	Lag the stack or measure other parts; useful as cross-checks

If you remember one thing: stage = median of the healthy TIR 4.6 stack probes, plus stack current. Heater and inlet-air signals only mark the exact switching minute.

4. 87 % idle, 13 % start-up, 0 % power

Stage - operation	Start	End	Min	T start	T end	°C/h	Heater max
Cold / idle - natural cooling	00:00	11:29	689	155.8	138.0	-1.6	0.0 kW
Start-up - air purge, heater off	11:29	12:27	58	139.5	116.3	-19.4	0.0 kW
Start-up - heat-up, heater on	12:27	14:31	124	116.1	215.5	+38.6	18.5 kW
Cold / idle - stopped, settling	14:31	16:10	99	199.1	198.5	0.0	0.1 kW
Cold / idle - natural cooling	16:10	24:00	470	198.5	181.1	-2.2	0.1 kW

Heat-up peaked at about 72 °C/h, well under the 120 °C/h limit in the guide. The heater ran at about 18 kW, less than half its 40 kW rating.

Questions for the operations team

• **Was 14:30 a trip or a planned stop?** Air and heater stopped in the same minute at only 216 °C. Please check the event log.
• **Is the 11:29 cooling a purge?** Air ran for an hour with the heater off before heating began.
• **What should "sitting warm, nothing on" be called?** It is labelled Cold / idle for now. The alternative is "Shut-down, natural cooling tail".

Files produced

`data_analysis/operating_states.py - the sorting script; run it with any day, for example: python data_analysis/operating_states.py 20260325`
`data_analysis/operating_states/20251030_states_1min.csv - every minute with stage, operation and signals`
`data_analysis/operating_states/20251030_segments.csv - the table above`
`data_analysis/operating_states/20251030_states.png - timeline chart`

If you remember one thing: the same script can now sort all 38 days. Stack current is recorded from 17 March 2026, so power stages can only show up from then on.

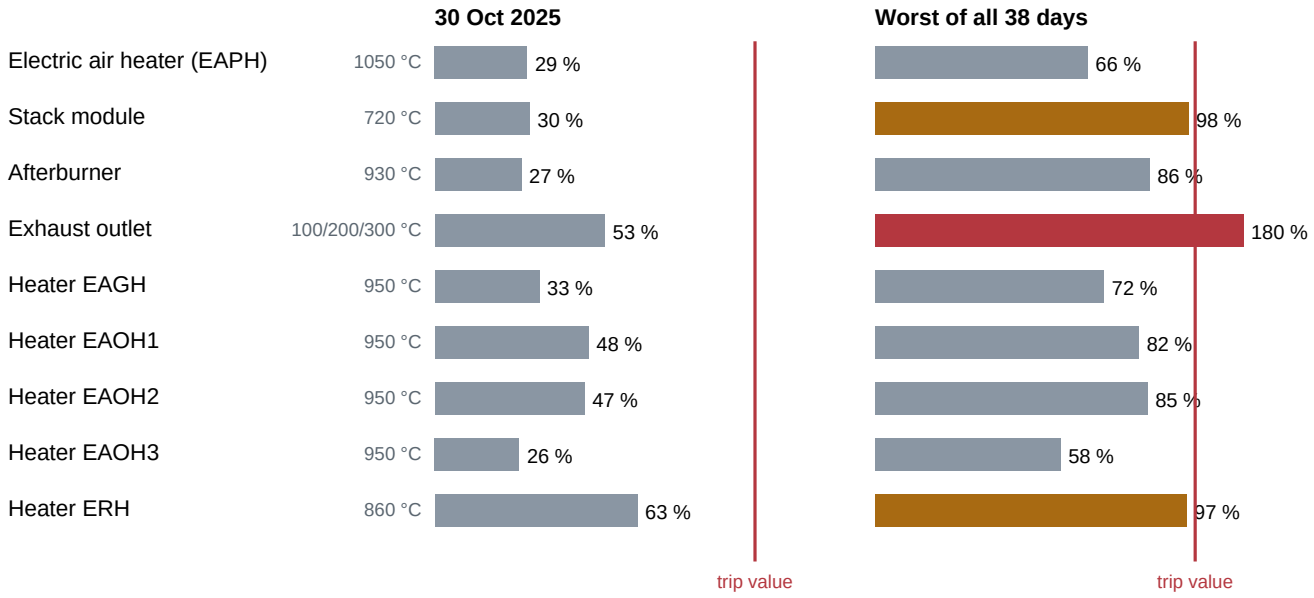
5. No safety trip on day one. The plant did trip in late March 2026, on the exhaust limit.

A **trip** is an automatic safety stop. A separate safety controller (the S-PLC) watches eleven limits. If any one is crossed for longer than 1/20 of a second, it shuts the plant down by itself.

Why it matters: a trip is a hard, sudden stop for a hot stack, and each one can age the cells. To judge wear later, we need to know which stops were trips and which were planned.

On 30 October 2025: every limit stayed far away. The closest was the exhaust outlet at 53 °C, about half of its 100 °C limit. The stop at 14:30 matches neither emergency pattern, so it was most likely a normal stop. Only the safety controller's event log can confirm that.

The eleven limits, and how close each came



Hottest reading of each safety sensor as a share of its trip value. Amber = within 5 % of the limit; red = went past it. The exhaust's 180 % is from November 2025 (limit then 100 °C), with the plant already off.

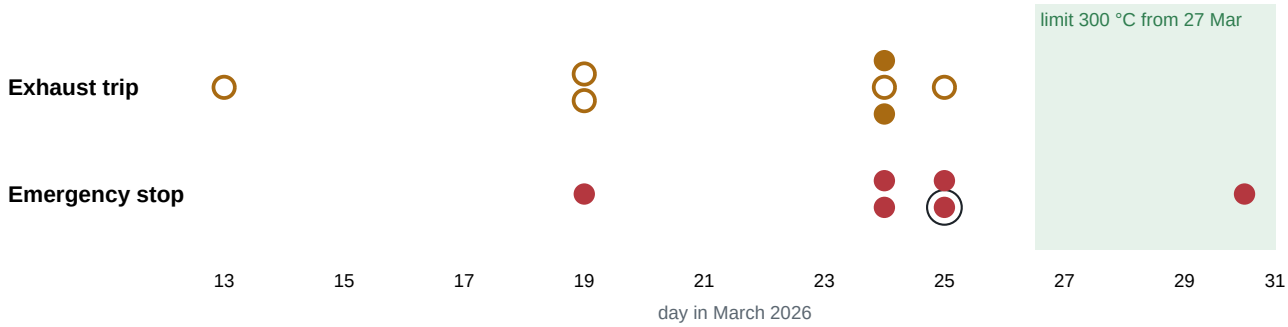
Other limit	Trips when	Day one	Worst, all days	Verdict
System pressure (PIRZ 8.1.2)	below 0.75 bar	1.00 bar	0.91 bar	Never tripped
Stack insulation (BE 1.1 - 5.1)	below 2 kOhm	not recorded	unit unclear	No trip seen; value falls through March, worth a look

Two kinds of emergency shutdown

The recorder does not log which trip fired. Each shutdown is recognised by what the plant did in the same second: which valves moved, whether power was cut, and whether the air compressor stopped.

	Emergency purge	Emergency stop
Set off by	a hot spot, low pressure, an insulation fault, or the exhaust fans	gas detector, smoke detector, stop button (none recorded)
What it does	shuts the fuel, opens flushing gas, cuts stack power and heaters at once	shuts every gas valve, opens one fuel vent, stops the air compressor
Found in the data	**2 complete + 5 where only power was cut**; all 7 seconds after the exhaust passed 200 °C	**6 times**, always with the stack hot (608 - 690 °C); cause not visible

When it happened: only in March 2026



Filled amber: full emergency purge. Open amber: exhaust passed 200 °C and power was cut, valves unchanged. Red: emergency stop (ringed: 25 Mar 08:50, cut 11.8 A of live load). No trip before 13 March; none on the exhaust after 27 March.

What this tells us

The exhaust limit caused the trips	Every time the exhaust passed 200 °C while running (7 of 7), power was cut 2 - 25 s later. The recorded value sat at 197 - 200 °C, so a short spike between two readings most likely set it off.
Six emergency stops are unexplained	One cut live load with no safety sensor near its limit. Three came 1 - 17 min after a purge trip and look like an operator reset sequence.
Fuel side left unprotected	After each emergency stop the hot fuel side went 4 - 10 min without protective gas, at 600 - 690 °C. Nine similar all-gas stops added 0.5 - 3 min each.
No equipment overheated	Tightest: stack module 16 °C below 720 °C (25 Mar), heater ERH 24 °C below 860 °C (13 Mar).

Questions for the operations team

#	Question	Needs
1	What set off the six emergency stops: gas detector, smoke, stop button, or an operator?	Event log
2	Why did five exhaust trips cut power without the valves moving?	Event log, settings
3	Does resetting after a purge trip pass through the emergency-stop state?	Procedure
4	Was 30 October 14:30 a planned stop?	Event log

If you remember one thing: day one had no trip. The plant's real trips came in late March 2026, set off by the 200 °C exhaust limit, and they stopped once that limit was raised to 300 °C.

Source: data_analysis/trip_check.py, notebook section 11; numbers in data_analysis/operating_states/trip_*.csv